1.6 The ε-δ Definition of a Limit

Name _____ Period ____ Date _____

AP Calculus 1.6 (30-minute plan) · Student sheet

I can…

- explain what ε and δ measure.
- find the largest δ that works for a given ε.
- write an ε-δ proof for the limit of a linear function.

Vocabulary · fill in the missing word

ε-δ definition of a limit — the limit of f(x) as $x \to c$ is L if for every $\varepsilon > 0$ there is a $\delta > 0$ so that $0 < |x - c| < \delta$ _____ $|f(x) - L| < \varepsilon$

Part 1 · Notes and examples (fill in each note, then work the example)

What they measure: ε is how close f(x) must be to _____; δ is how close x must be to c.

Example 1

The limit of $2x$ as $x \to 1$ is 2. Find the largest δ that works for $\varepsilon = 0.01$.

Definition: for every $\varepsilon > 0$ there is $\delta > 0$ so that $0 < |x - c| < \delta$ implies $|f(x) - L| <$ _____.

Linear functions: for $f(x) = mx + b$, $m \neq 0$, $\delta =$ _____ works.

Example 2

$f(x) = 4x - 1$ and the limit as $x \to 3$ is 11. Find δ for $\varepsilon = 0.1$, for $\varepsilon = 0.01$, and for any $\varepsilon > 0$.

Writing the proof: "Given $\varepsilon > 0$, let $\delta = \ldots$" and end by showing $|f(x) - L|$ _____ ε.

Example 3

Prove that the limit of $2x + 5$ as $x \to 0$ is 5.

Part 2 · Partner practice

Partner practice

Prove that the limit of 3x as x → 3 is 9.